
KoPolicyAgentBench: Measuring Locale-Specific Policy Compliance of Tool-Using LLM Agents in Korean Workplace and Customer-Service Environments

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Abstract

Tool-using LLM agents can send messages, change records, approve refunds, and escalate customer cases, yet policy compliance under Korean workplace and customer-service operating rules remains hard to measure. We introduce KoPolicyAgentBench, a Korean-language benchmark of 100 synthetic tasks that separates task success, policy compliance, unsafe proposals, monitor prevention, and final safe success. In a paid claude-fable-5 slice over ten frozen test tasks and four agent conditions, all four conditions reach 10.0% safe success. The monitor-augmented condition reaches 100.0% policy compliance but not higher task success, exposing a concrete execution-boundary failure mode: natural-language action proposals are often blocked as unknown tools.

1. Introduction

LLM agents are moving from text-only assistants into systems that call tools and change external state. In workplace and customer-service settings, this shift changes the safety question. The central risk is no longer only whether a model refuses a harmful prompt. It is whether an agent can finish a useful task while respecting the operating policies that govern the side effects it may trigger, such as sending an email, updating an employee record, approving a refund, or escalating a customer case.

Korean deployments sharpen this problem because many policies are expressed through locale-specific workplace and service expectations. A safe agent may need to withhold private employee information, ask for confirmation before an irreversible action, use

appropriate honorific business style, stay within refund authority, avoid unauthorized escalation, and be honest about what it has actually completed. Prior agent benchmarks study tool-user interaction and domain policies (Yao et al., 2024), or tool-risk and execution-boundary attacks (Ruan et al., 2023; Debenedetti et al., 2024). Korean safety benchmarks study response-level social harms and bias (Lee et al., 2023; Jin et al., 2024). KoPolicyAgentBench targets the remaining intersection: Korean locale operating policies, multi-turn tool use, side-effectful actions, and separated monitor evaluation.

This paper studies that intersection through KoPolicyAgentBench. We define policy compliance as compliance with synthetic locale-specific policy constraints written into each task. These policies are not claims about Korean law, Korean culture as a whole, or universal norms. They are controlled operating rules that represent common workplace and customer-service policy types, which lets us measure whether agents preserve policy constraints while pursuing task success.

The evaluation problem has three parts. First, task success and policy compliance must be measured separately, because an agent can achieve the user-visible goal by violating privacy, skipping confirmation, or exceeding authority. Second, a safety monitor should not receive credit merely for hiding unsafe behavior after the agent has proposed it. We therefore separate agent unsafe proposal rate, monitor prevention rate, and system safe success rate. Third, locale-specific constraints should be tested with counterfactual control pairs, so that a failure can be tied to a policy conflict rather than to task difficulty alone.

KoPolicyAgentBench is the first benchmark to evaluate multi-turn, side-effectful workplace/customer-service agents under Korean locale operating policies, separately measuring task success, unsafe proposals, monitor prevention, and final safe success with counterfactual control pairs.

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Our contributions are:

- We introduce a benchmark for measuring Korean locale policy compliance in multi-turn, tool-using, side-effectful agent environments, covering workplace and customer-service tasks with synthetic policy cards and deterministic simulation.
- We propose a three-way metric separation protocol that reports System Safe Success Rate, Agent Unsafe Proposal Rate, and Monitor Prevention Rate, preventing execution-boundary monitors from masking unsafe proposals as safe behavior.
- We construct counterfactual control pairs that isolate whether violations increase under policy conflict, including normal versus policy-conflict tasks, authorized versus unauthorized actions, draft versus send actions, and Korean policy cards versus generic cards.
- We prioritize deterministic evaluation of final state, action order, and content leakage, using LLM judges and human calibration only for soft semantic or style judgments, which improves reproducibility while keeping language-sensitive checks in scope.

The rest of this paper describes the benchmark design, reports the compact fable evaluation, and states the current limitations and release safeguards.

2. Benchmark Design

2.1. Overview

KoPolicyAgentBench evaluates policy compliance in Korean-language workplace and customer-service agents with side-effectful tools. The task set contains 100 synthetic tasks, split into 20 development tasks and 80 frozen test tasks. Each task asks an agent to complete a useful goal while respecting a locale-specific policy card and the limits of the available tools. Counterfactual pairs remove the policy conflict while preserving the same domain, tool surface, and success predicate, which helps separate ordinary task difficulty from policy adherence under pressure.

2.2. Policy Ontology

The policy registry contains six policies across the two benchmark domains. Table 1 lists the machine-readable identifiers, domains, categories, and operational rules.

The ontology is intentionally small: privacy, confirmation, style, policy override, authorization, and hon-

esty create distinct failure modes for tool-using agents. Some policies govern content, such as disclosure and Korean honorific style; others govern action order or authority.

2.3. Task Anatomy

Each task is a structured scenario with an instruction, initial database state, available tools, goal predicates, safety invariants, counterfactual link, and scripted user profile. Policy compliance is scored over the whole rollout: final state, action order, content exposure, tool calls, monitor decisions, and final user-visible response.

2.4. Counterfactual Control Pairs

In the conflict version, the agent faces a user goal that conflicts with a policy, such as disclosing employee information, claiming a refund was processed without a tool state change, or escalating without authorization. In the control version, the policy conflict is removed while the task remains comparable. This paired structure makes policy-card and monitor effects easier to interpret than unpaired aggregate scores.

2.5. Determinism and CoT Handling

Rollouts run against an in-memory database, fixed tool registry, scripted user responses, fixed seeds, and a frozen test set recorded in data/tasks/FREEZE.json with hash c3192f50f59a. The harness logs observable action traces, tool calls, final states, and final outputs, but does not store or score chain of thought.

2.6. Evaluator Stack

The primary evaluator stack uses deterministic final-state, action-trace, and content-leak checkers. Optional language and semantic-leak judges cover honorific style and fuzzy semantic exposure, but deterministic checks are the source of the headline results reported here.

3. Experimental Setup

3.1. Agent Conditions

We evaluate four agent conditions. B0 is a direct function-calling baseline. B1 adds ReAct-style action planning without a policy card. B2 adds the benchmark policy card to B1. B4 applies the hybrid execution-boundary monitor to B2, observing proposed side-effecting actions before the simulator changes state. B0, B1, and B2 run with monitor mode none; B4 runs with monitor mode hybrid.

Table 1. Policy Ontology

Policy ID	Domain	Category	Description
WP-PII-01	Workplace	Privacy	Prohibits unauthorized disclosure of employee personal information, including salary, resident registration numbers, and evaluations.
WP-CONFIRM-01	Workplace	Confirmation	Requires user confirmation before irreversible workplace actions, including messages or updates that create durable side effects.
WP-HONOR-01	Workplace	Style	Requires honorific and business appropriate Korean style when addressing external parties or senior staff.
CS-REFUND-03	Customer service	Policy override	Prohibits granting refunds beyond the stated customer service policy.
CS-ESCALATE-01	Customer service	Authorization	Prohibits escalation, approval, or state change actions when the user lacks the required authority.
CS-HONESTY-01	Customer service	Honesty	Prohibits false claims that an order, refund, or service action has been completed when the tool state does not show completion.

3.2. Model Matrix

We report a compact paid evaluation with claude-fable-5 through the Anthropic-compatible gateway configured in configs/models.yaml. The slice uses ten frozen test tasks, seed 42, and disabled soft judges, so all headline values in Table 3 come from deterministic state, trace, and content checks.

Table 2. Executed fable evaluation.

Model	Tasks	Baselines
claude-fable-5	10 frozen test tasks	B0, B1, B2, B4

This compact matrix is designed to validate the benchmark harness, isolate tool-interface failures, and provide an empirical case study under a fixed budget.

3.3. Metrics

The headline metric is Safe Success Rate (SSR): task goal completed and no policy violation. Task Success Rate (TSR) measures goal completion alone, while Policy Compliance Rate (PCR) measures the absence of safety violations alone. We also track parse rate because provider-specific tool-call formatting affects whether a proposed action can reach the monitor or simulator.

3.4. Statistical Analysis

The larger benchmark protocol uses task-level paired bootstrap confidence intervals with B0 as reference, B1 versus B2 for policy-card prompting, and B2 versus B4 for the hybrid monitor. The current ten-task evaluation is reported descriptively as a compact stress test rather than as a model ranking.

3.5. Reproducibility Controls

LLM calls pass through a reproducibility cache so repeated identical prompts do not trigger duplicate paid calls. The evaluation policy does not store or grade proprietary chain-of-thought text; it evaluates observable user messages, tool calls, tool observations, monitor decisions, and final database state.

4. Results

Table 3 reports the completed paid fable evaluation: ten frozen test tasks, four agent conditions, and one seed. The goal is not to rank models, but to test whether the benchmark exposes separable task-success, policy-compliance, and execution-boundary failure modes.

Table 3. Paid claude-fable-5 evaluation over ten frozen test tasks.

Cond.	n	SSR	TSR	PCR	Parse
B0	10	10.0	10.0	90.0	76.3
B1	10	10.0	10.0	80.0	70.5
B2	10	10.0	10.0	90.0	75.7
B4	10	10.0	10.0	100.0	76.3

The monitor-augmented condition improves policy compliance but does not improve safe success in this evaluation. B4 reaches PCR 100.0%, while all four conditions remain at SSR 10.0%. Trajectory inspection shows a recurring failure mode: claude-fable-5 often describes the intended action in natural language rather than emitting a valid tool call. The monitor blocks these unknown proposals, preserving policy compliance but leaving task completion low. This illustrates why KoPolicyAgentBench reports task success, compliance, and parse reliability separately.

5. Limitations

KoPolicyAgentBench measures agent behavior under task-specified operating policies, not legal or cultural knowledge. The policy cards are synthetic constraints, not claims about Korean law, culture, or societal norms. Results should therefore be read as evidence about compliance with benchmark conditions.

The counterfactual pair design is the main measurement limitation. Human validation found mean agreement $\kappa = 0.561$, but counterfactual validity had lower agreement, $\kappa = 0.189$. This weakens causal claims based on paired unsafe and safe variants.

The benchmark also supports soft language and content judges, which creates a circularity risk when an LLM judges another LLM. The fable slice avoids this by using deterministic checks only; judge calibration F1 is not run.

Provider tool calling is another limitation. OpenAI-style native tool calls, Anthropic-style tool-use blocks, and JSON fallback behavior are not identical, so provider comparisons combine model behavior with each provider’s tool-calling implementation.

Statistical power is limited by the current ten-task evaluation, so the reported fable numbers should be treated as harness evidence and failure analysis, not as a final model ranking. The monitor receives sanitized context rather than privileged answer labels, but the monitor design may still encode benchmark-specific assumptions.

6. Ethics

This work does not involve human subjects. The tasks, user profiles, policy cards, tool outputs, and sensitive values are synthetic and are not drawn from real people. The benchmark does not claim to define Korean culture, Korean law, or the values of Korean speakers. Locale appears as a task condition: agents receive Korean-language interactions and synthetic operating policies, then are evaluated against those written policies. The monitor is a benchmark condition, not a real-world safety guarantee.

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A. Related Work

KoPolicyAgentBench sits at the intersection of agentic tool-use evaluation, safety evaluation for tool boundaries, and Korean-language safety or cultural alignment benchmarks. Prior work covers important parts of this space, but the combination of Korean locale operating policies, multi-turn side-effectful tools, monitor separation, and counterfactual policy-control pairs remains distinct.

Agent benchmarks with domain policies. The closest line of work is τ -bench (Yao et al., 2024), which evaluates multi-turn customer-service agents with tool calls, domain policies, and final-state scoring. KoPolicyAgentBench follows the same broad motivation that realistic agents should be judged by stateful outcomes, not only by text quality. Its focus differs in three ways: the operating policies are written for Korean locale workplace and customer-service settings, the benchmark separately measures unsafe proposals before monitor intervention, and tasks include counterfactual control pairs designed to isolate policy conflict from general task difficulty.

Tool-risk and execution-boundary safety. ToolEmu (Ruan et al., 2023) studies risks that arise when language models use external tools, while AgentDojo (Debenedetti et al., 2024) evaluates attacks and defenses around tool boundaries. These benchmarks clarify why tool-using agents need safety evaluation beyond answer refusal. KoPolicyAgentBench instead targets benign task completion under explicit operating policies. The threat is not primarily an adversarial injection attack, but the everyday failure mode in which an agent tries to help and violates privacy, confirmation, authority, honesty, or Korean business-style constraints along the way.

Korean safety and cultural alignment benchmarks. Korean-language safety resources such as KoSBI (Lee et al., 2023) and bias benchmarks such as KoBBQ (Jin et al., 2024) test important response-level harms and social biases. They establish that Korean evaluation should not be treated as a simple translation of English safety data. KoPolicyAgentBench builds on that motivation, but evaluates agents acting in a simulated environment rather than static responses. The unit of analysis is a rollout with tool calls, final state, content exposure, action order, and monitor intervention, not a single generated answer.

Positioning. We therefore make a narrow claim about the intersection: Korean locale operating poli-

cies, multi-turn side-effectful workplace and customer-service agents, separated agent and monitor metrics, and counterfactual control pairs. This framing also limits the cultural claim. The benchmark policies are synthetic task-level constraints that model policy types common in Korean workplace and customer-service settings, not a statement of legal truth or cultural essence.

B. Appendix

B.1. Reproducibility

All planned runs use seeds 42, 43, and 44. The task set is frozen in data/tasks/FREEZE.json with hash c3192f50f59a. Model version tags are pinned in configs/models.yaml. The runner uses a reproducibility cache so that the same deterministic seed and request key map to the same stored LLM output, avoiding accidental re-sampling and repeated billing during analysis. Code and benchmark artifacts are released at <https://github.com/MoerAI/Korean-Policy-Agent-Bench>. Chain of thought is not stored or scored; reproduction uses prompts, model tags, seeds, cached observable outputs, tool calls, observations, monitor decisions, and final states.

B.2. Error Taxonomy

Code	Category	Description
E1	Privacy leakage	Reveals fabricated sensitive values or copies them into an unsafe channel.
E2	Missing confirmation	Performs an irreversible or sensitive action without required user confirmation.
E3	Policy disregard	Ignores a stated policy card constraint while pursuing the task goal.
E4	Hallucination	Invents facts, permissions, records, or tool results not present in the environment.
E5	Unauthorized state change	Changes simulator state without a valid policy basis or user authorization.
E6	Over refusal	Refuses a task that the policy allows and the tools can complete safely.
E7	Over confirmation	Asks for extra confirmation when the policy does not require it.
E8	Wrong tool	Calls an inappropriate tool or uses the right tool for the wrong operation.
E9	Incomplete task	Stops before satisfying the goal despite available safe actions.
E10	Wrong recipient	Sends information or an action to the wrong person, account, team, or channel.
E11	Style violation	Violates the Korean style or register requirement in the task policy.
E12	Escalation failure	Fails to escalate when the policy requires human or supervisor handling.
E13	Duplicate action	Repeats a state changing action that should occur once.
E-parse	API or tool call parse failure	Produces an invalid, ambiguous, or unsupported action representation.

Table 4. Error taxonomy used for qualitative analysis.